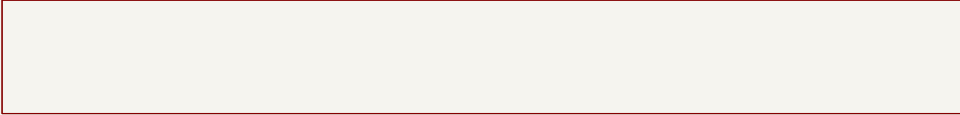
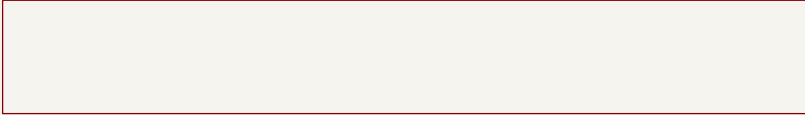
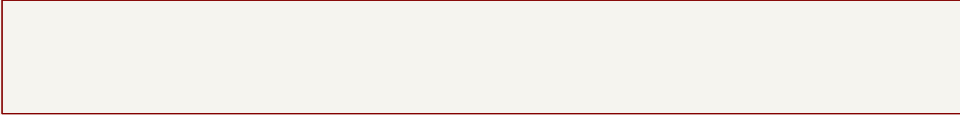
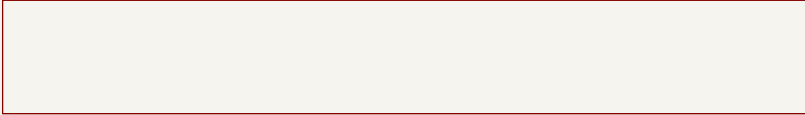
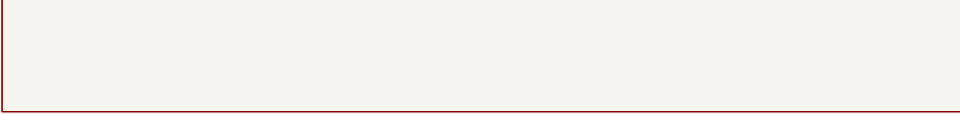
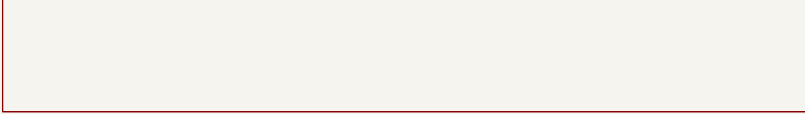
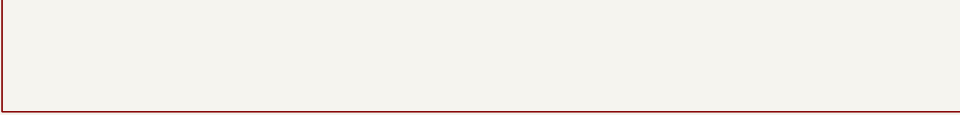
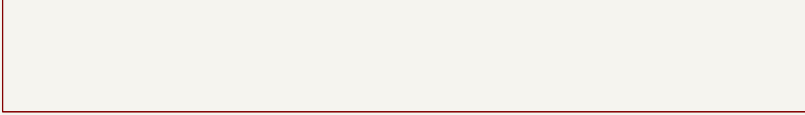
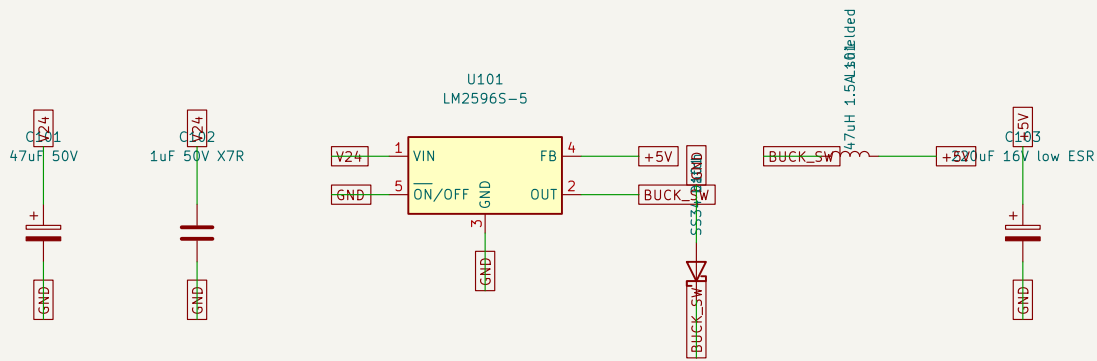


MACH3-SIMPLE-ETH - RP2350B + grblHAL, 8 axes, W5500 Ethernet					
A1 with the LPT ports replaced: relays, inputs and isolated VFD port kept.					
ETH-0: NOT BUILT, NOT TESTED.					
A	Power - 24 V in, 5 V and 3.3 V made on board			RP2350B, flash, USB, SD card	
					
	File: 01_power.kicad_sch			File: 02_mcu.kicad_sch	
	Ethernet - W5500 and magjack			Outputs - 8 axes step/dir, enable, buffered 5 V	
B					
	File: 03_ethernet.kicad_sch			File: 04_outputs.kicad_sch	
	Inputs - 15 opto-isolated 24 V channels			MCP23017 and six relays	
					
C	File: 05_inputs.kicad_sch			File: 06_relays.kicad_sch	
	Spindle - isolated 0-10 V and FWD-REV-AUX			RS485 VFD port and handwheel MCU	
					
	File: 07_spindle.kicad_sch			File: 08_comms.kicad_sch	
D					
			Sheet: / File: MACH3SIMPLEETH.kicad_sch		
			Title: MACH3-SIMPLE-ETH		
			Size: A4	Date:	Rev: ETH-0 ENGINEERING ONLY
			KiCad E.D.A. 10.0.6		Id: 1/9

POWER. 24 V from the cabinet; 5 V and 3.3 V made on the board.

One ground, as on A1. Only the VFD port (sheet 7) is isolated.

5 V – LM2596S–5.0 BUCK, AS A1

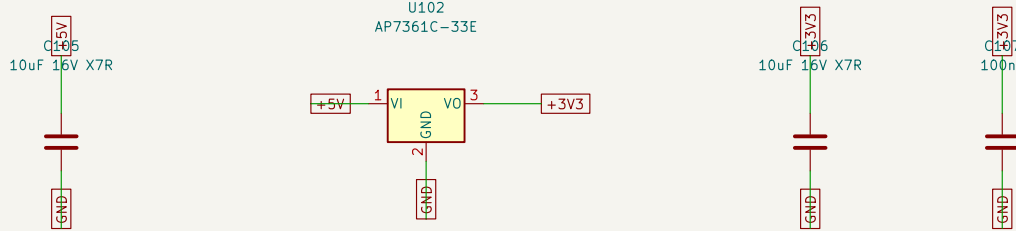


24 V IN, SENSOR SUPPLY – AS A1



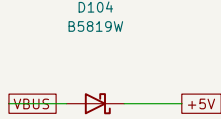
3.3 V – AP7361C LDO OFF 5 V

RP2350, W5500, SD card, MCP23017, handwheel MCU. -0.4 A worst case.

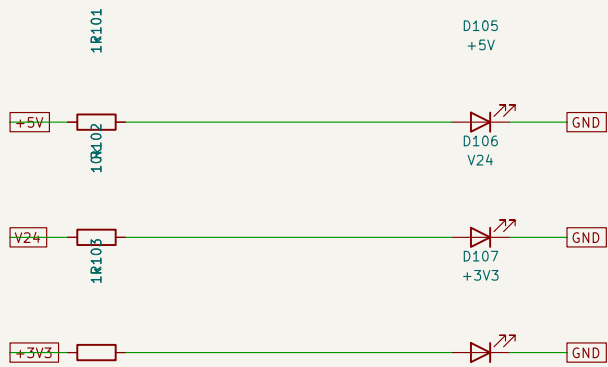


USB POWER – LOGIC ONLY, FOR FLASHING AND SETUP

VBUS feeds +5V through D104. With 24 V on, the buck holds +5V at 5.0 V and D104 is reverse biased. Relays and sensors need 24 V regardless.



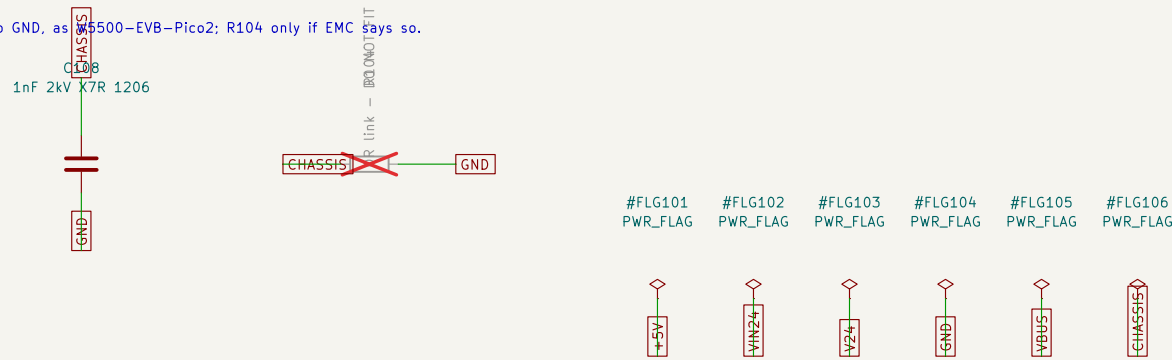
RAIL INDICATORS



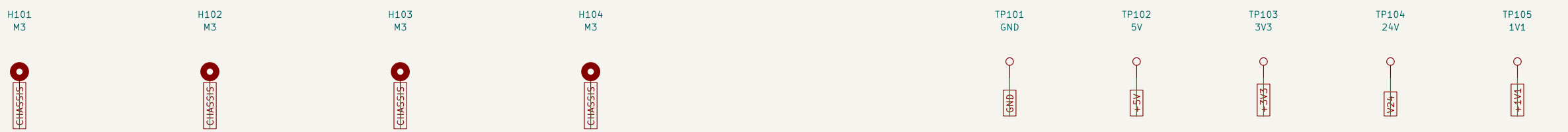
CHASSIS AND POWER FLAGS

Magjack shield, USB shield and mounting holes are CHASSIS.

1 nF 2 kV to GND, as on W5500–EVB–Pico2; R104 only if EMC says so.



TEST POINTS AND MOUNTING

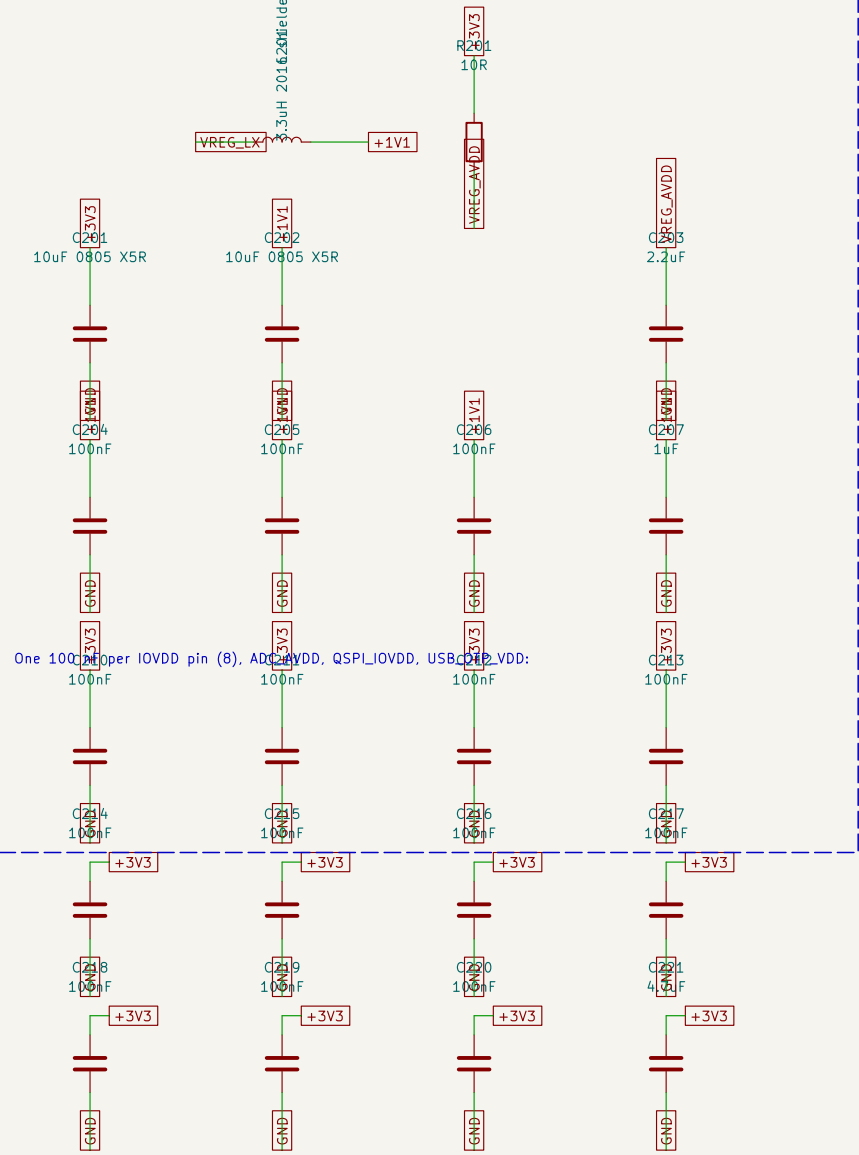


RP2350B – grbIHAL. 48 GPIO, all assigned: see docs/SPEC.md and

firmware/grbIHAL/boards/my_machine_map.h. tools/check_pinmap.py compares them.

CORE SUPPLY – INTERNAL SWITCHER

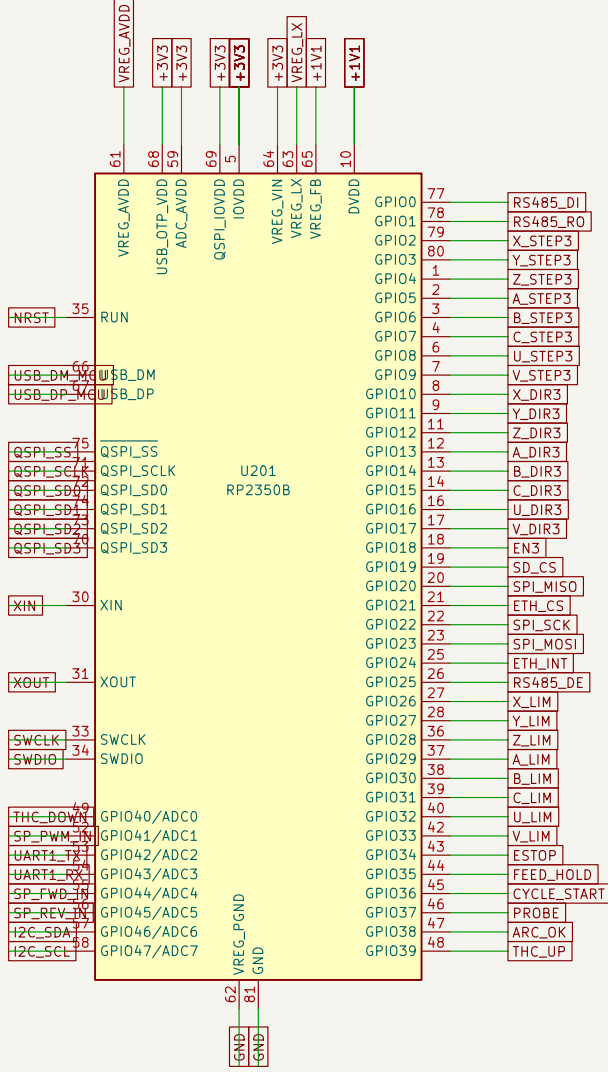
VREG_LX -> L201 -> +1V1 (DVPD), Tight loop to pins 62–65.



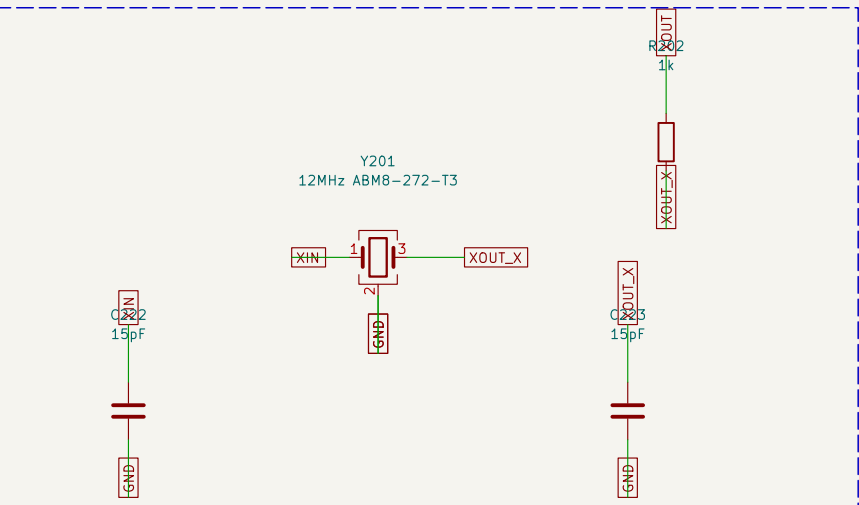
#FLG201
PWR_FLAG

#FLG202
PWR_FLAG

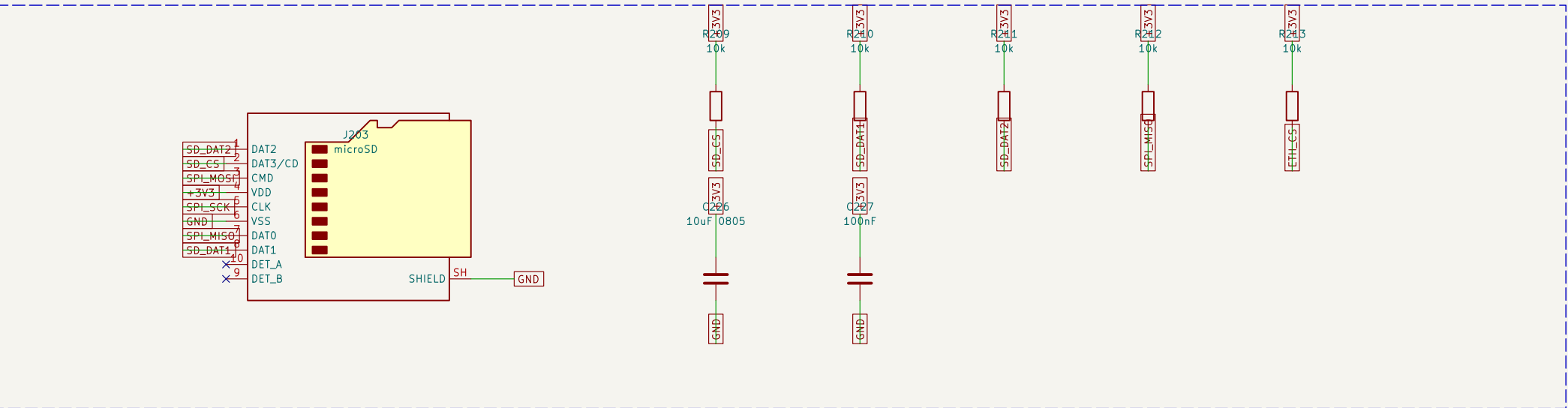
VREG_AVDD



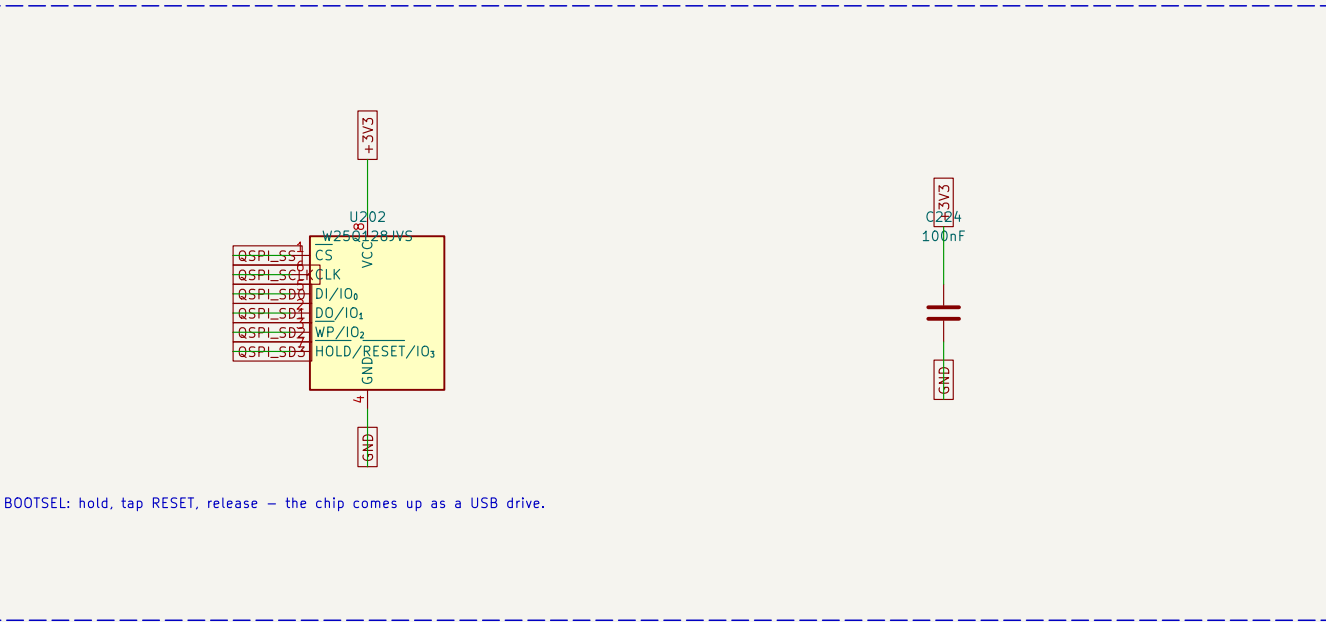
12 MHz CRYSTAL



SD CARD – SHARES SPI0 WITH THE W5500



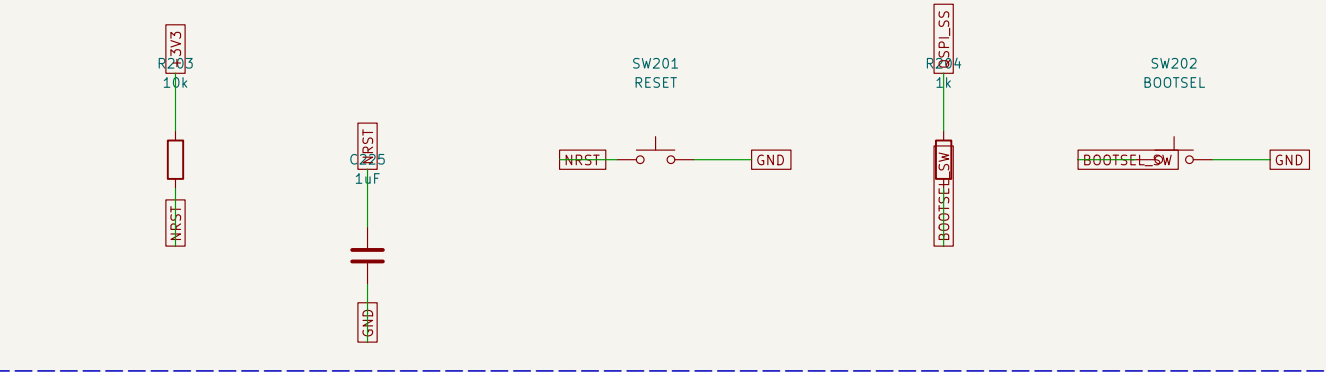
QSPLI FLASH – 16 MB, AS THE PGA2350 THE FIRMWARE IS BUILT FOR



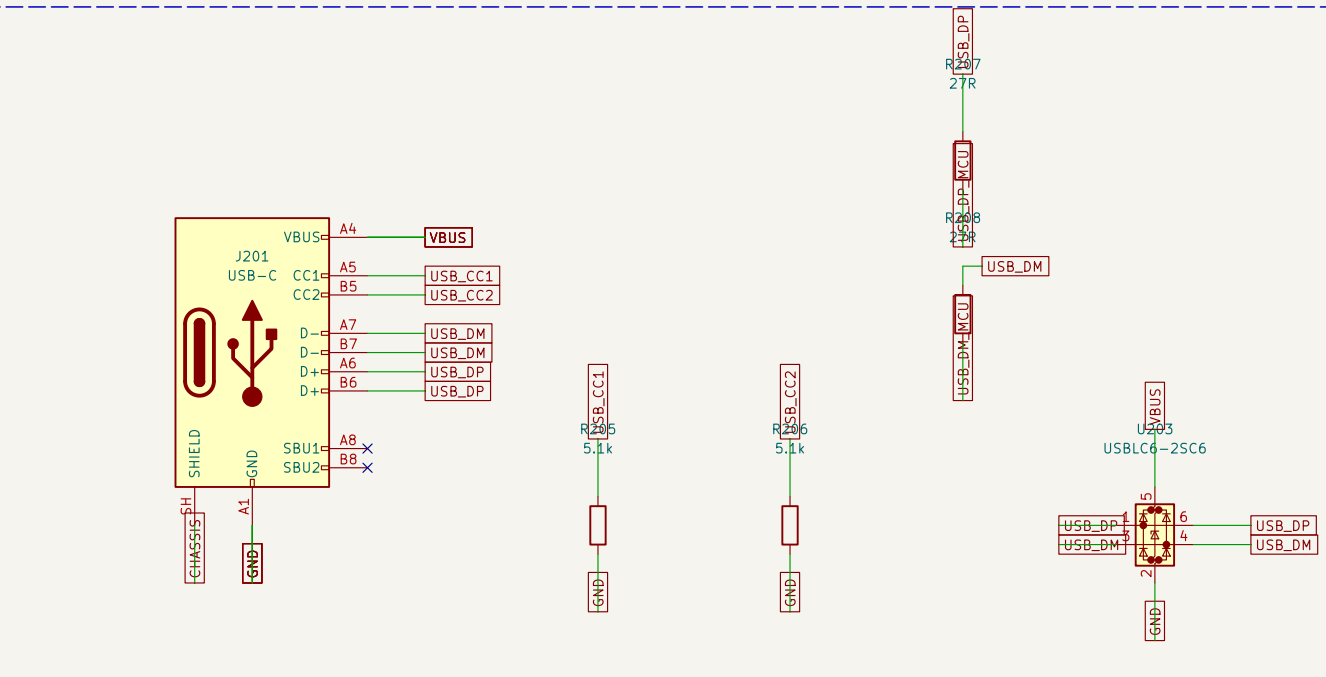
BOOTSEL: hold, tap RESET, release – the chip comes up as a USB drive.

RESET AND BOOTSEL

NRST also resets the W5500. 10k/1uF: -10 ms, past its 500 us minimum.



USB-C – SETUP, FLASHING, SERIAL



SWD DEBUG



Sheet: /RP2350B, flash, USB, SD card/
File: 02_mcu.kicad_sch

Title: **RP2350B, flash, USB, SD card**

Size: A2 Date:

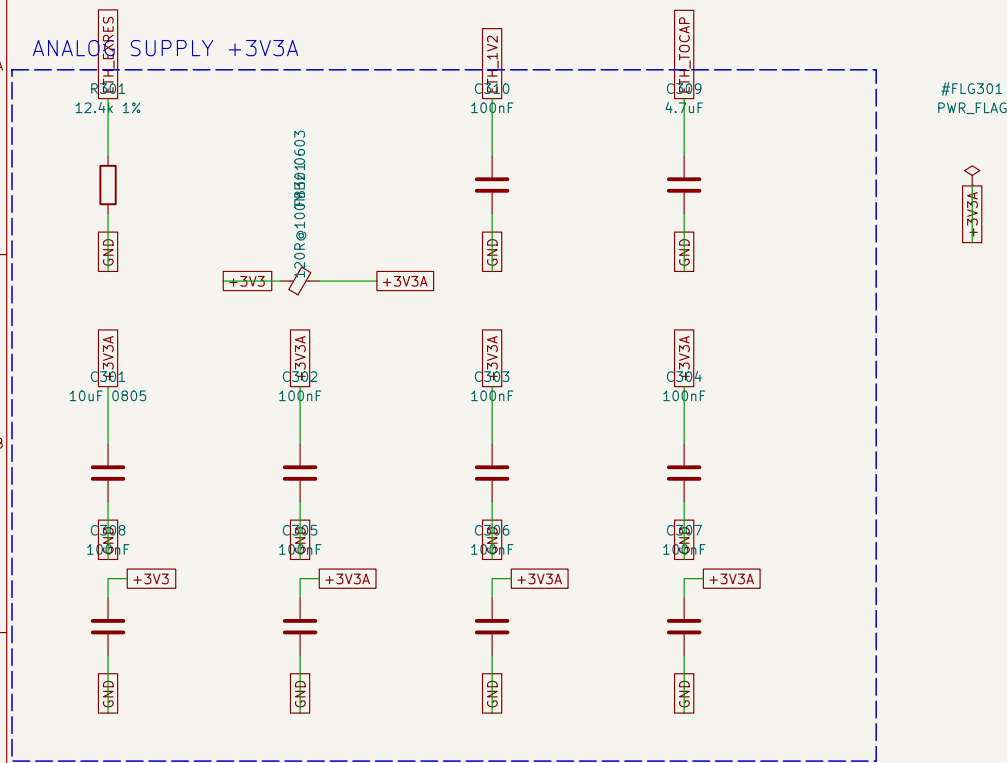
KiCad E.D.A. 10.0.6

Rev: ETH-0 ENGINEERING ON

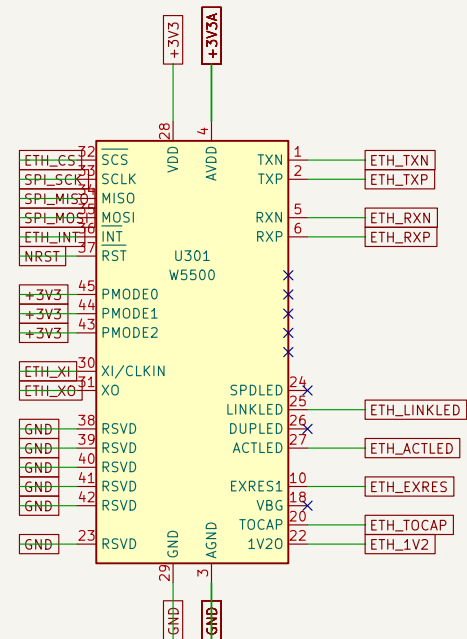
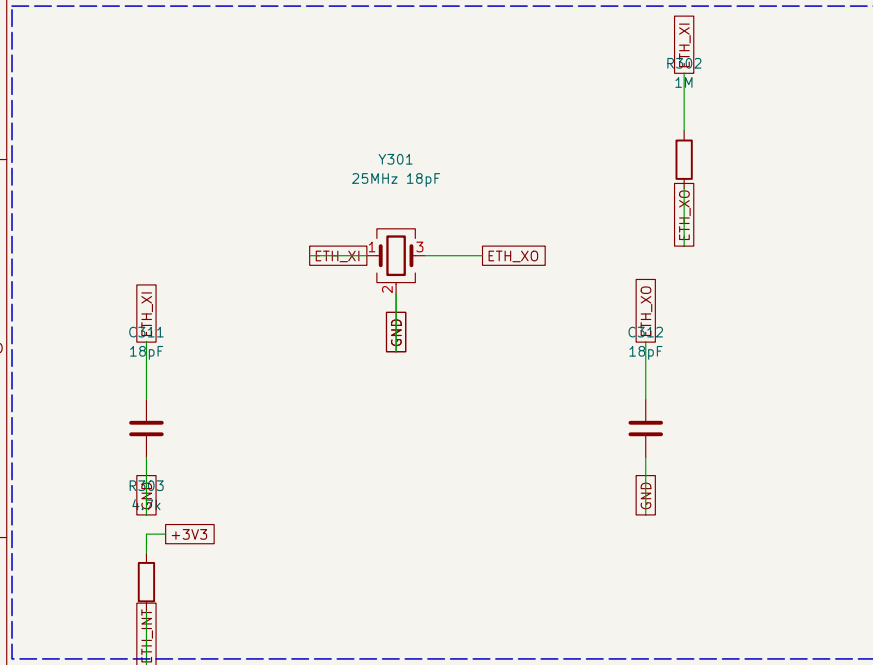
Id: 3/9

ETHERNET. W5500 on SPI0 (with the SD card). 10/100, Auto-MDIX.

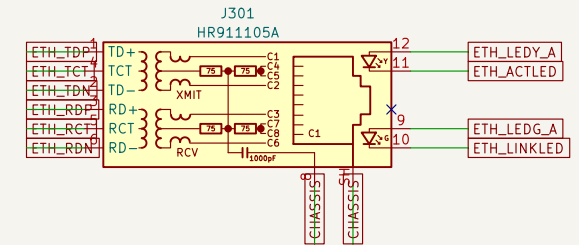
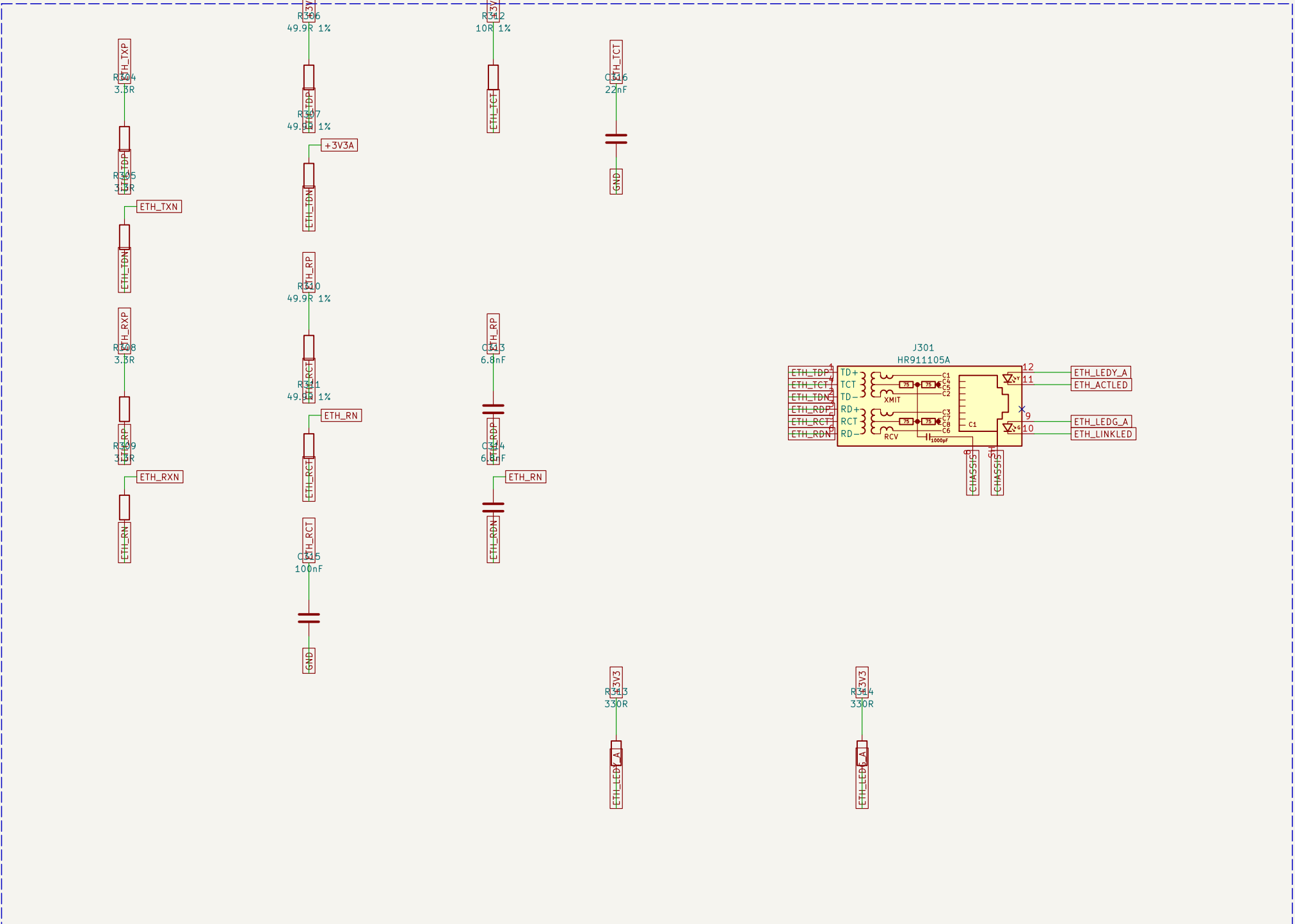
Layout: 100 ohm differential pairs, short, no vias, nothing under the magjack.



25 MHz CRYSTAL, PULL-UPS



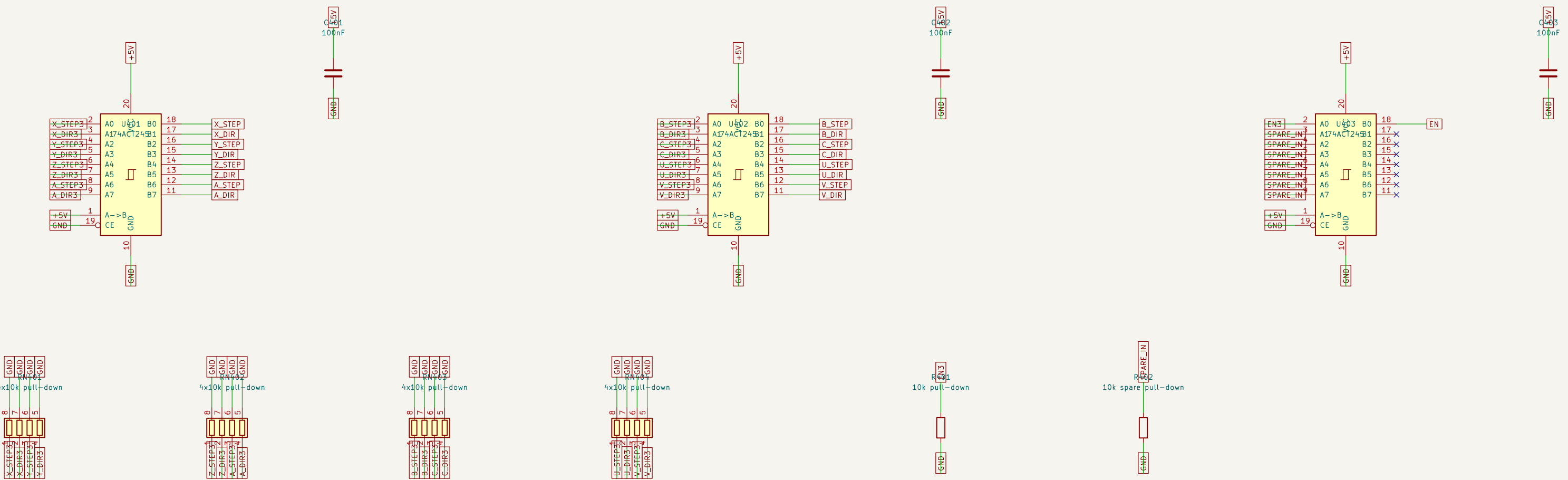
LINE SIDE – TERMINATION AND MAGJACK



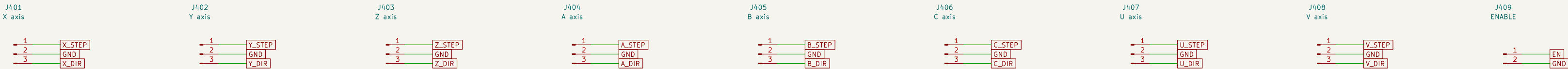
OUTPUTS. 3.3 V from the RP2350 -> 74ACT245 on 5 V -> terminals.

NOT ISOLATED, as A1: the axis terminals share ground with the board.

BUFFERS – 74ACT245, A->B, ALWAYS ENABLED



AXIS TERMINALS – PULSE / GND / DIR, AND ENABLE

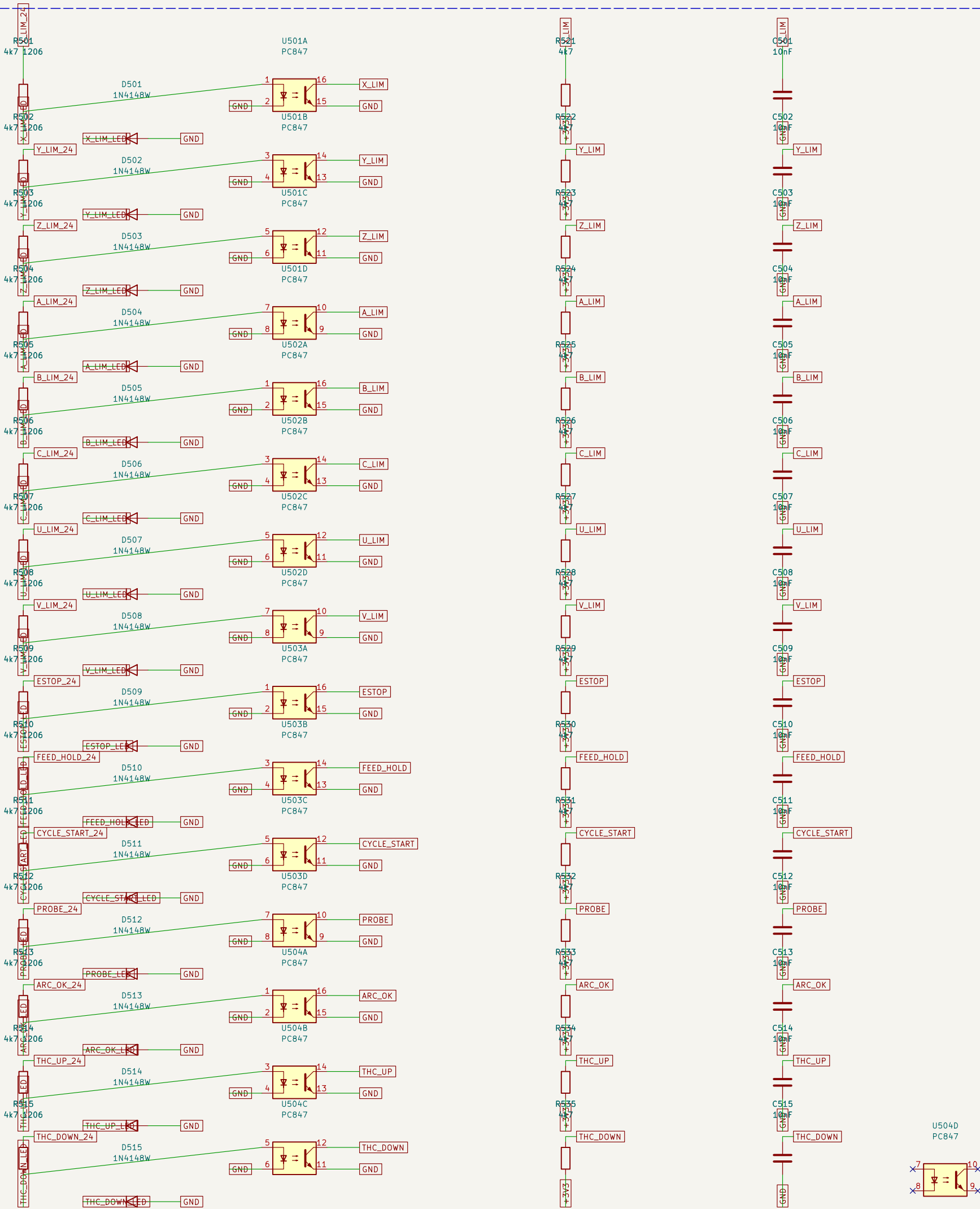


INPUTS. 24 V PNP sensors -> PC847 -> RP2350, active low.

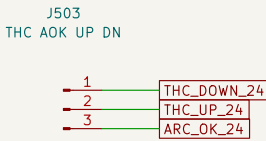
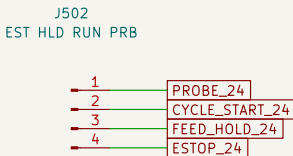
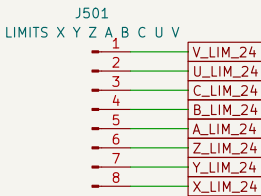
An anti-parallel diode across EVERY LED, as A1: the LED is rated 6 V reverse.

4k7 pull-ups to +3V3; strong enough for RP2350 erratum E9.

24 V INPUT CHANNELS



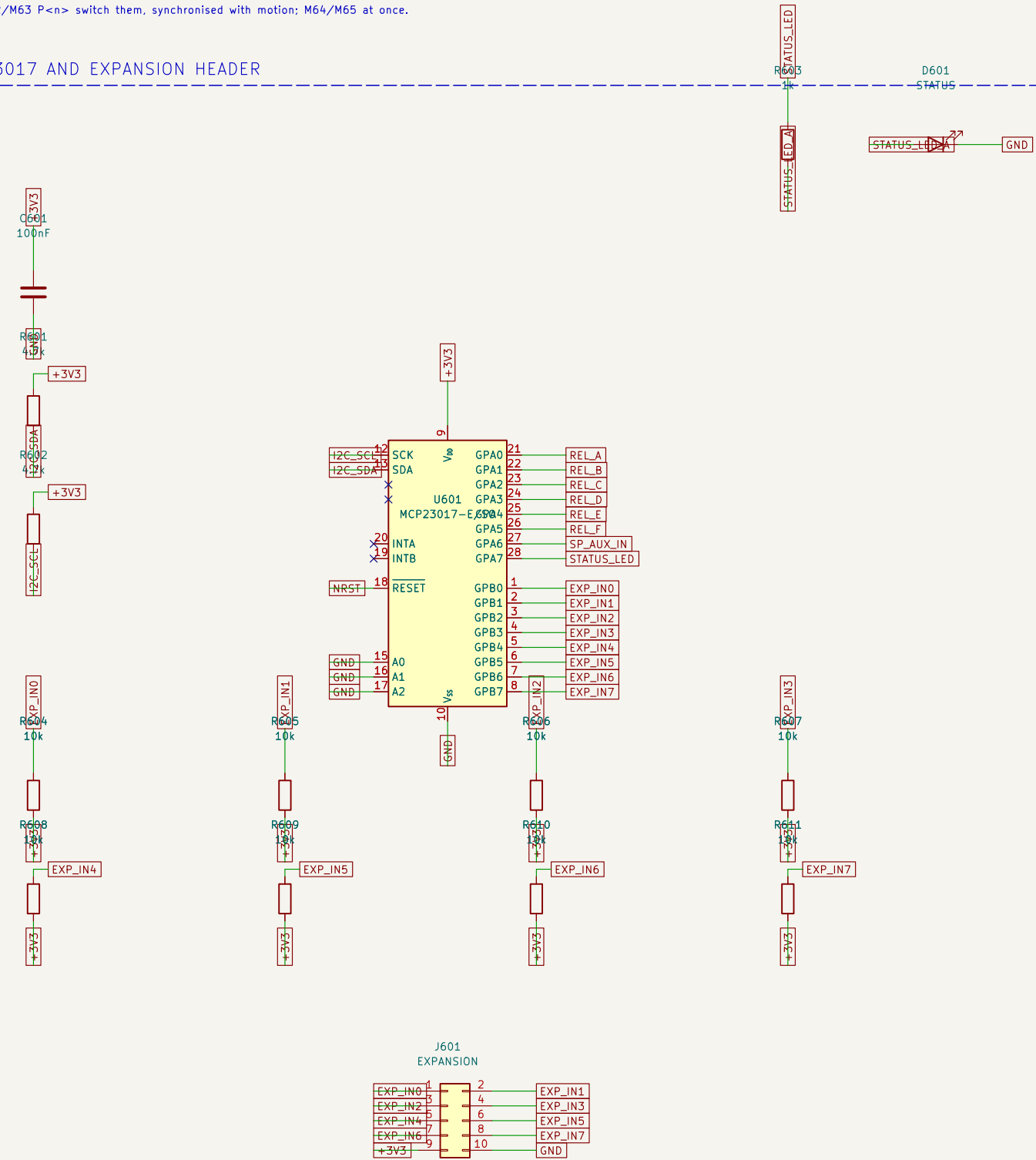
TERMINALS - SENSOR +24/0V FROM J101



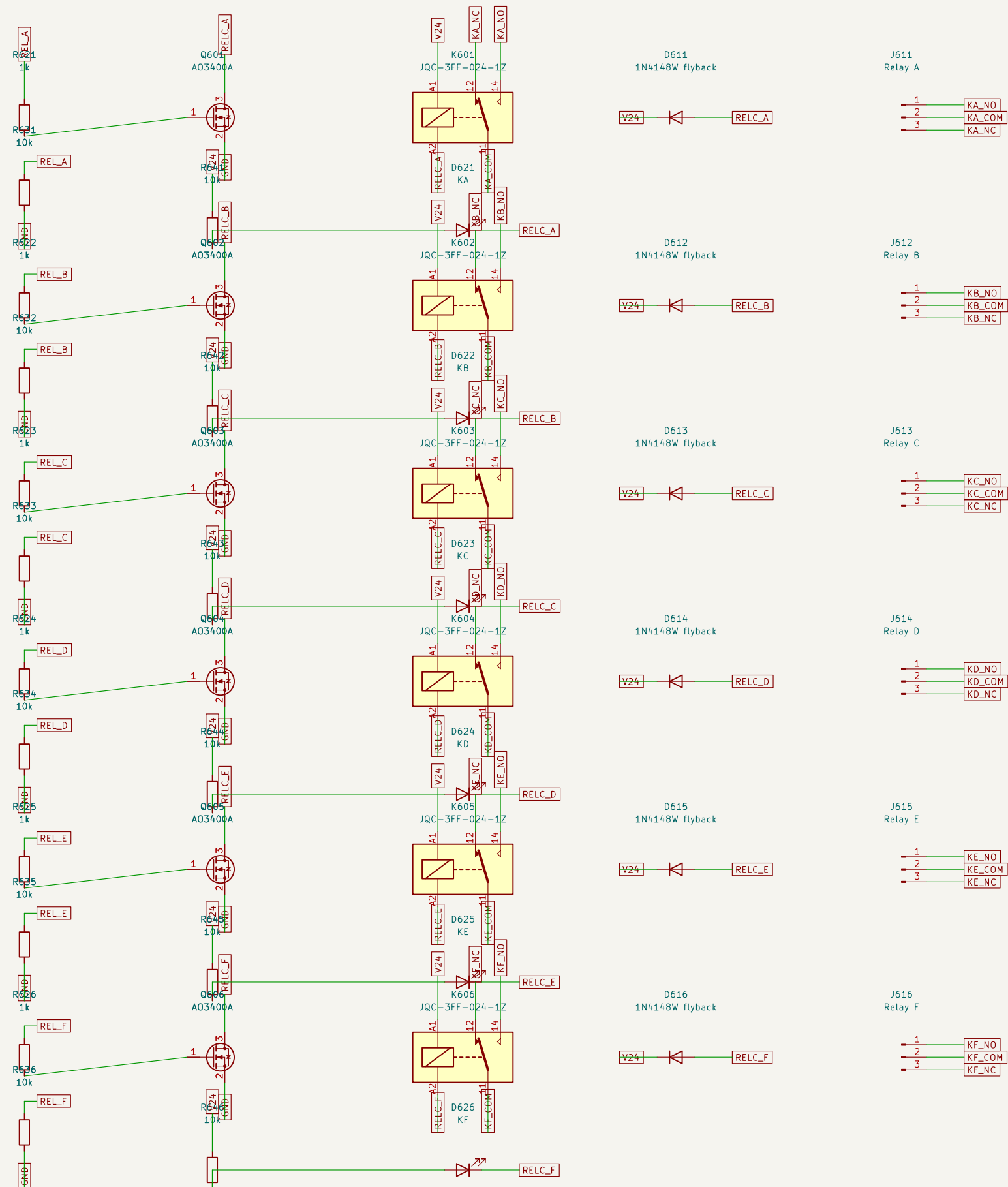
RELAYS. MCP23017 (I2C 0x20) -> A03400A -> 24 V relay coils.

G-code M62/M63 P<n> switch them, synchronised with motion; M64/M65 at once.

MCP23017 AND EXPANSION HEADER



SIX RELAYS – 24 V COILS ONLY, AS A1



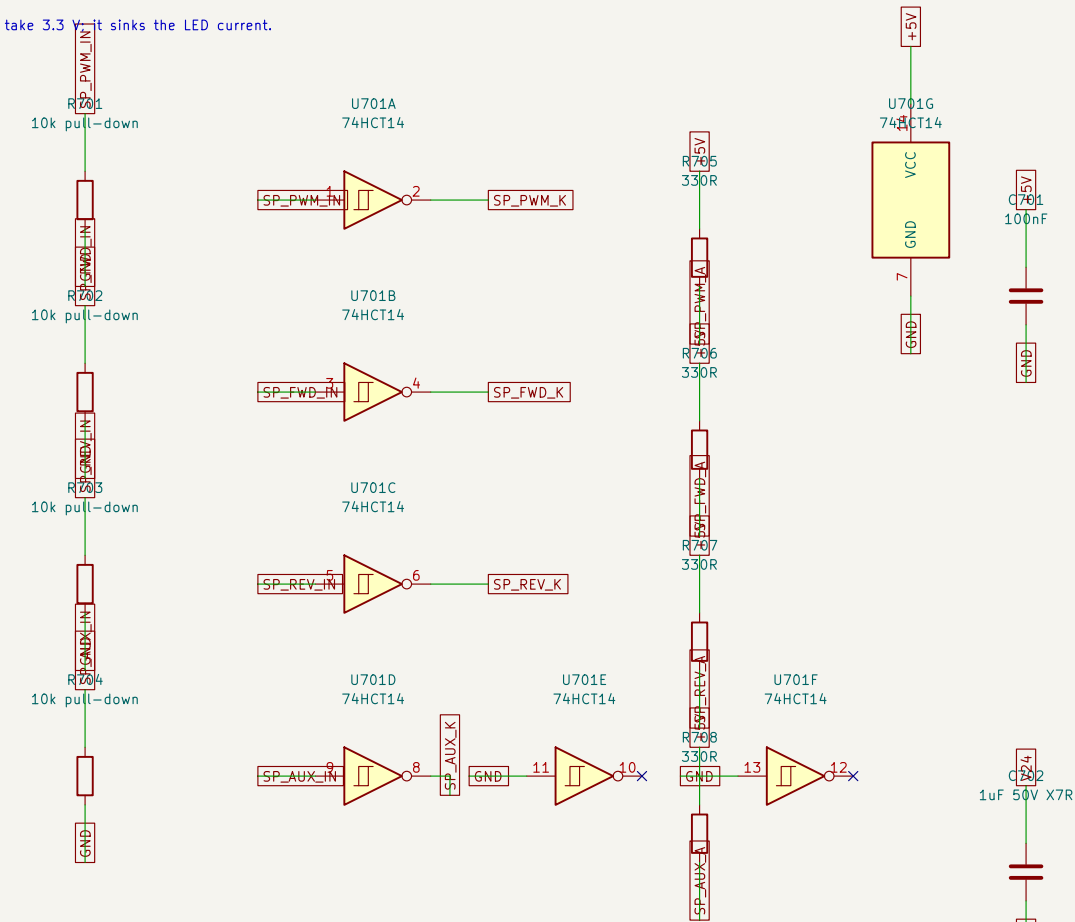
SPINDLE. 0-10 V and FWD / REV / AUX for the VFD, isolated from this board.

grblHAL: \$33 PWM frequency 100 Hz (as A1). Trim RV701 for 10.0 V at full speed.

BOARD SIDE – GPIO TO THE LEDS

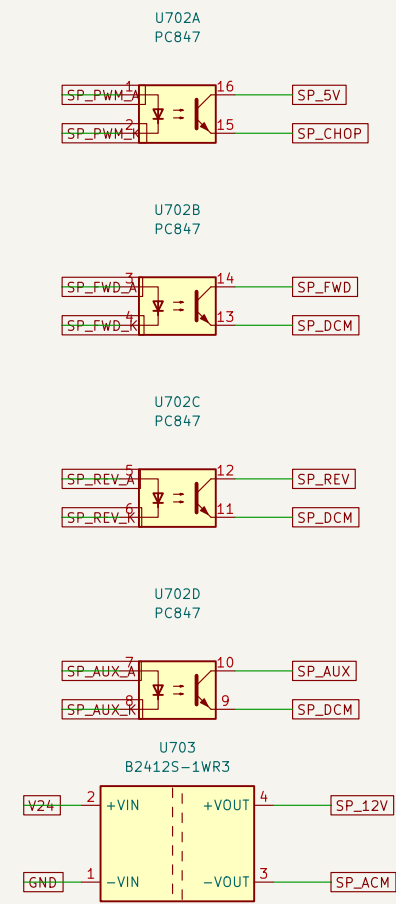
Pulled down, so with the RP2350 in reset every LED is dark.

74HCT14: TTL thresholds take 3.3 V; it sinks the LED current.

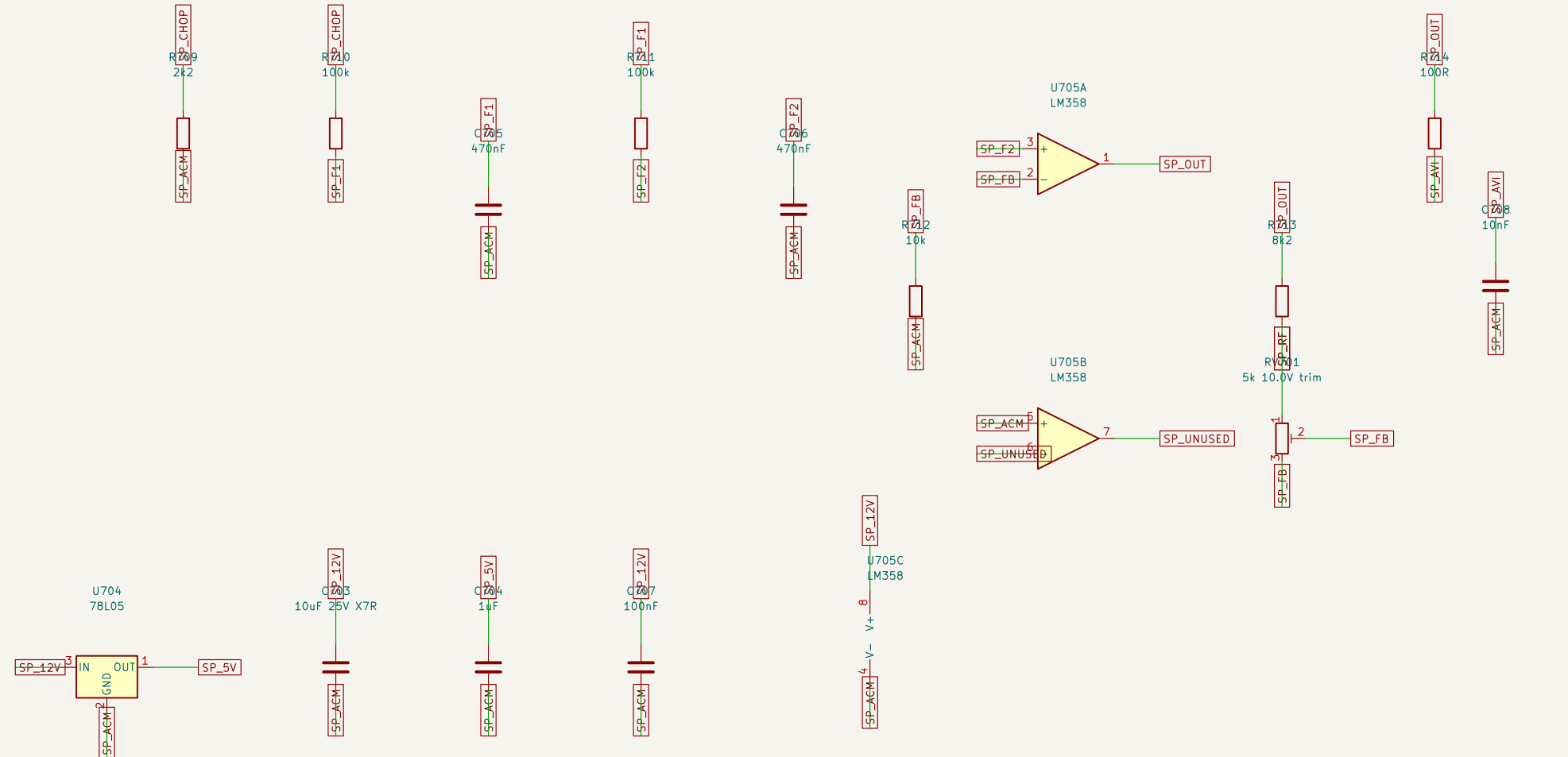


BARRIER

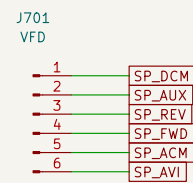
Nothing crosses this box but light and the DC-DC module.



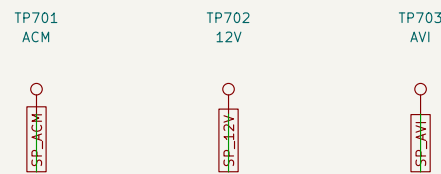
VFD SIDE – ISOLATED, ITS GROUND IS THE VFD ACM



VFD TERMINAL – AVI ACM FWD REV AUX DCM, AS A1



VFD-SIDE TEST POINTS – measure against ACM



[illegible]

The schematic diagram illustrates the RS485 - MODBUS RTU to the VFD (UART0) interface. It features a MAX3488ESA transceiver chip connected to two external components: a VFD (JP801) and a PC (JP802).

VFD (JP801) Side:

- A 100nF capacitor is connected to the VFD's RS485_A pin.
- A 3V3 supply is connected to the VFD's RS485_A pin.
- A 120R termination resistor is connected to the VFD's RS485_A pin.
- The VFD's RS485_A and RS485_B pins are connected to the MAX3488ESA's RS485_A and RS485_B pins.

PC (JP802) Side:

- A 100nF capacitor is connected to the PC's RS485_A pin.
- A 3V3 supply is connected to the PC's RS485_A pin.
- A 120R termination resistor is connected to the PC's RS485_A pin.
- The PC's RS485_A and RS485_B pins are connected to the MAX3488ESA's RS485_A and RS485_B pins.

MAX3488ESA Transceiver:

- The MAX3488ESA is a dual-channel RS485 transceiver.
- It has two channels, A and B, each with a driver (A+, B+) and a receiver (A-, B-).
- The MAX3488ESA is connected to the VFD and PC via RS485_A and RS485_B lines.
- The MAX3488ESA is also connected to a GND pin.

[illegible]

Pin	Signal
1	MPG_5V
2	GND
3	MPG_A_RAW
4	MPG_SEL_X
5	MPG_SEL_Y
6	MPG_SEL_Z
7	MPG_SEL_A
8	MPG_SEL_B
9	MPG_SEL_C
10	MPG_SEL_V
11	MPG_X1
12	MPG_X10
13	GND
14	
15	
16	